

Data Cleaning & Preparation

The dataset was structurally clean and required minimal preprocessing. Data quality validation was performed to check for missing values, duplicate records, invalid dates, inconsistent text formatting, and datatype issues. No significant data quality problems were identified, and the dataset was found to be ready for analysis.

To support more effective business analysis and dashboard development, additional helper columns were created:

- **Order Year** – enabled year-wise sales and profitability trend analysis.
- **Order Month** – enabled month-wise trend analysis and dashboard filtering.
- **Month Number** – ensured chronological month sorting (Jan → Dec) instead of alphabetical ordering.
- **Profit Margin %** – enabled profitability analysis by comparing profit relative to sales.
- **Profit/Loss Flag** – classified transactions into profit-making or loss-making orders for performance analysis.
- **Discount Bucket** – grouped discount levels (No, Low, Medium, High) to analyze the relationship between discounting and profitability.